

SQL COURSE FOR BEGINNERS - PART 1

- **From Database Fundamentals to SQL**
 - **Beginner-Friendly**
 - **PostgreSQL**
 - **Relational Databases**
 - **SQL Queries**
 - **Database Design**
- **What You'll Learn**
 - **Database Fundamentals**
 - **Relational vs. Non-Relational Databases**
 - **DBMS & RDBMS**
 - **PostgreSQL**
 - **Databases & Tables**
 - **CRUD Operations**
 - **SQL Queries**
 - **Table Relationships**
 - **Database Schemas**
 -



WHAT IS A DATABASE?

WHAT IS A DATABASE (DB)?

- Definition
 - Any organized collection of related information.
- Everyday Examples
 - Smartphone contact list
 - WhatsApp chat history
 - To-do list on your phone
 - Credit card statement
- Storage Methods
 - Physical (notebook, paper)
 - Mental (memory)
 - Digital (computer files, cloud storage)

WHY WE NEED DIGITAL DATABASES

High-Volume Systems

- Examples: Digital Bank/Fintech
- Stores millions of user accounts, balances, and transactions.
- Requires 24/7 uptime and instantaneous data access.
- Critical data: errors lead to massive financial loss.
- High Security: Mandatory protection for sensitive data (tax IDs, passwords, encryption keys).

vs

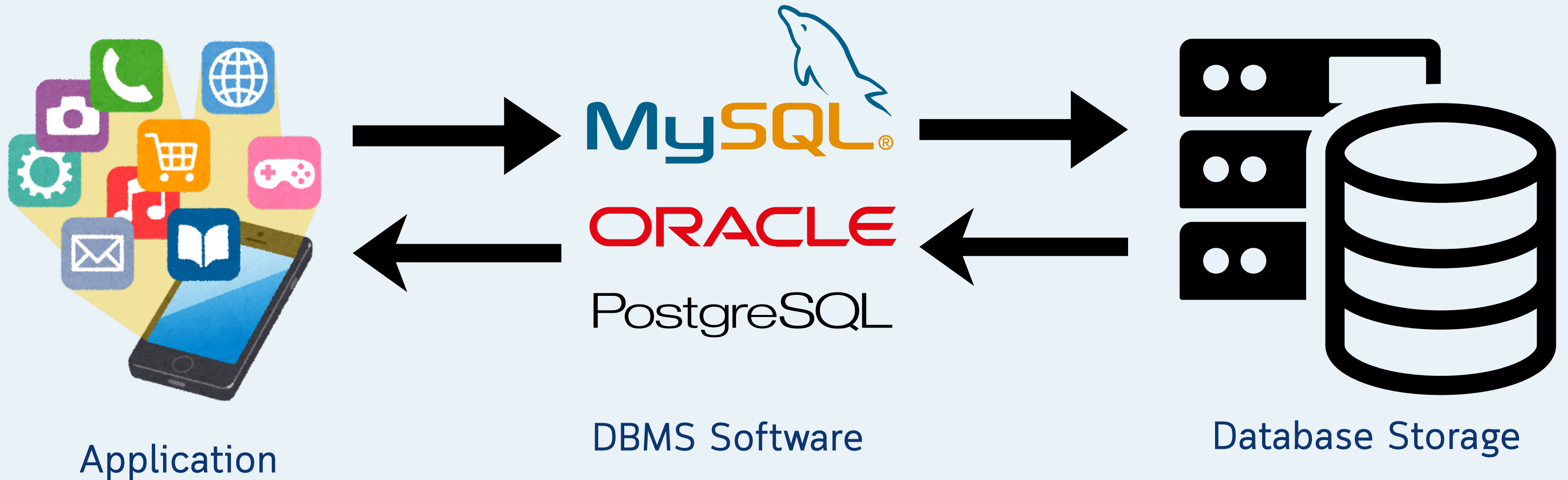
Personal Notes

- Examples: Notepad App
- Stores 10 to 50 simple notes.
- Casual, local access only.
- Convenient, but non-critical data.
- Low or no security needed.

WHAT IS A DATABASE MANAGEMENT SYSTEM (DBMS)?

- DBMS Definition
 - Specialized software designed to create, manage, and secure databases.
- Core Responsibilities
 - Security & Access Control
 - Manages user permissions for reading and editing data.
 - Backups & Recovery
 - Prevents data loss during system failures.
 - Concurrency Control
 - Allows multiple users to interact with data simultaneously without conflict.
 - Interface
 - Connects backend software (Python, Java, Node.js) to the database.

HOW APPLICATIONS INTERACT WITH A DBMS



- The application sends requests to read or modify data.
- The DBMS validates security, processes the request, and executes the operation on the storage layer.

The image features decorative blue geometric patterns in the corners. The top-left corner has thin, curved lines. The top-right corner has a dark blue shape with white geometric lines. The bottom-left corner has a dark blue shape with white geometric lines. The bottom-right corner has thin, curved lines.

C.R.U.D

CREATE

READ

UPDATE

DELETE

TWO TYPES OF DATABASES

- Relational Database (SQL)
 - Organize data into one or more tables
 - Each table has columns and rows
 - A unique key identifies each row
- Non-Relational Database (noSQL)
 - Organize data is anything but a traditional table
 - Key-value store
 - Documents (JSON, XML, etc)
 - Graphs
 - Flexible Tables

RELATIONAL DATABASE (SQL)

Clients Table

client_id (PK)	full_name	ssn_tax_id
101	Sarah	111-222
102	Marcus	333-444
103	Ellen	555-666
104	Rippley	777-888

Bank Accounts Table

account_id (PK)	client_id	balance
5001	101	2500.50
5002	102	10400.00
5003	103	150.00
5004	104	21400.00

THE SQL ECOSYSTEM & RDBMS

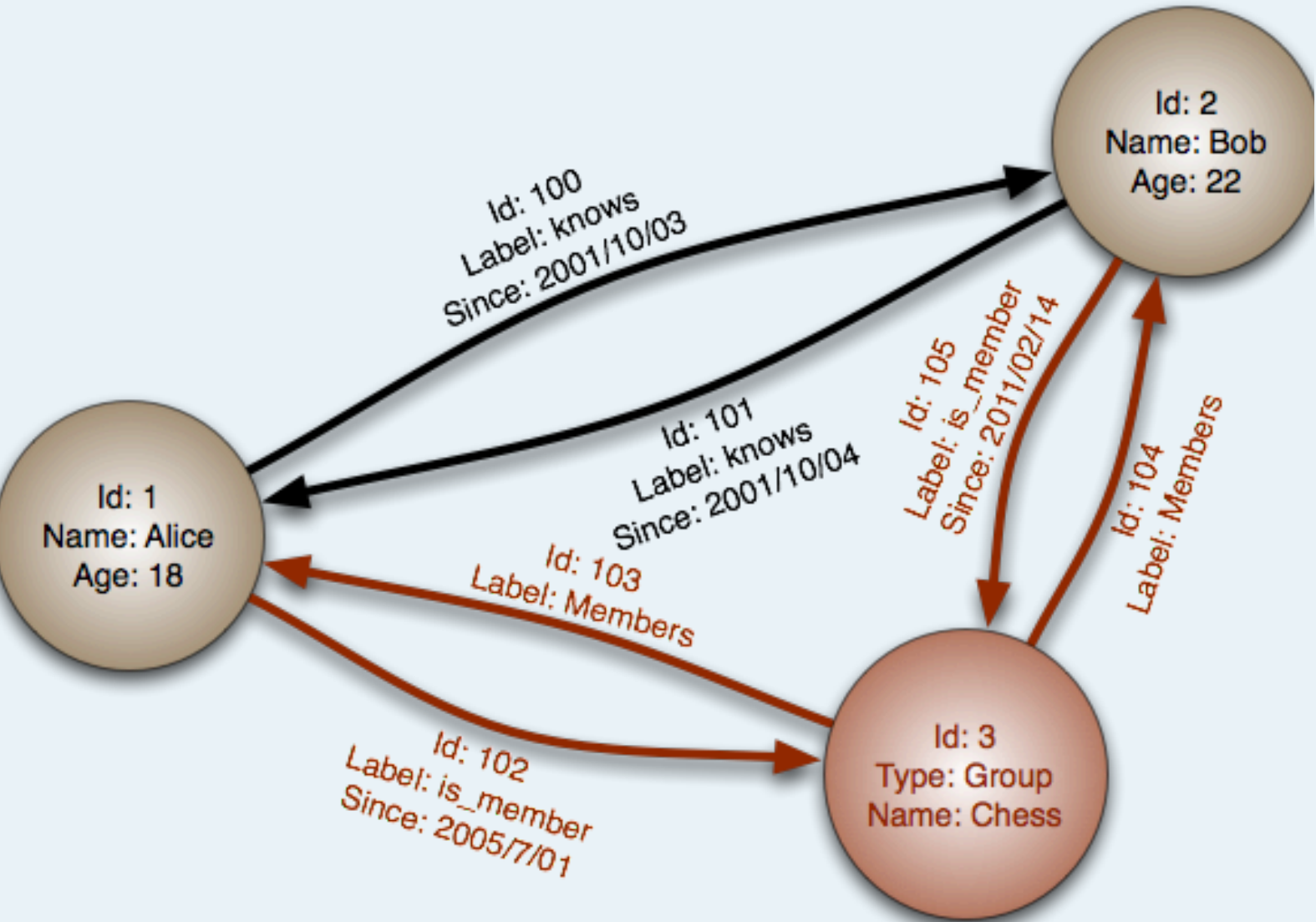
- RDBMS (Relational Database Management System):
 - Popular software: PostgreSQL, MySQL, Oracle, SQL Server, MariaDB.
- SQL (Structured Query Language):
 - Standardized language used to communicate with an RDBMS.
 - Executes CRUD operations, manages permissions, and defines schema structures.

NOSQL DATABASE MODELS

```
1 [
2   {
3     "Id": 1,
4     "Name": "John",
5     "Age": 24
6   },
7   {
8     "Id": 2,
9     "Name": "Paul",
10    "Age": 19
11  },
12  {
13    "Id": 3,
14    "Name": "Roger",
15    "Age": 21
16  }
17 ]
```

DOCUMENT

(JSON, XML, BLOB, ETC...)



GRAPH

RELATIONAL NODES

KEY	HASH	ANGLE (DEG)
"john"	1633428562	58.7
"C"	2269549488	81.7
"kate"	3421657995	123.1
"jane"	5000799124	180
"A"	5572014557	200.5
"bill"	7594634739	273.4
"B"	8077113361	290.7
"steve"	787173343	352.3

KEY-VALUE HASH

KEYS ARE MAPPED TO
VALUES (STRINGS, JSONS,
BLOB, ETC...)

THE NOSQL ECOSYSTEM

- Popular NoSQL DBMS Platforms
 - MongoDB, DynamoDB, Redis, Cassandra, Firebase.
- No Universal Standard
 - Each engine relies on its own custom query API or syntax.
 - Schema-less flexibility allows data structures to evolve without rigid migrations.

WHAT ARE DATABASE QUERIES?

- Definition
 - A formal request sent to a database system to retrieve or manipulate specific data.
- Complexity
 - As databases grow and relationships multiply, queries require higher structural logic.
- Real-world Analogy
 - Running a Google search or applying filters on an online banking dashboard are examples of queries.

WRAP UP

- Database
 - Any organized collection of related information.
- DBMS
 - Software that creates, manages, and secures databases.
- CRUD
 - Create, Read, Update, Delete operations.
- Relational (SQL)
 - Organizes data into tables with primary/foreign key relationships.
- Non-Relational (NoSQL)
 - Stores data using alternative structures (Documents, Key-Value, Graphs).
- Queries
 - Instructions used to retrieve and manipulate stored data.